Strong Law for IID Variables with a Finite Fourth Moment

An End-to-End Meta-Math Walkthrough

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Abstract

The fourth moment supplies a summable n^{-2} deviation bound. A countable collection of tolerances then upgrades eventual control to almost sure convergence. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PS-02 / Probability and Stochastic Processes / Report v1

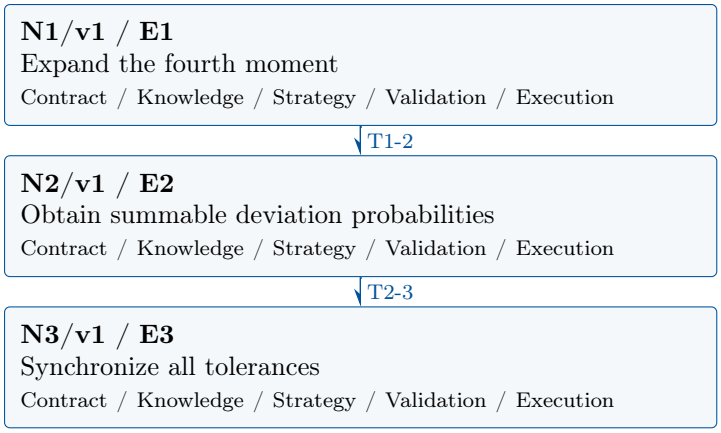
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $(X_n)_{n\geq 1}$ be independent identically distributed real random variables with $\mathbb{E}|X_1|^4 < \infty$ and mean μ. Prove $n^{-1}\sum_{j=1}^{n} X_j \to \mu$ almost surely. This is the finite-fourth-moment version, not the optimal integrability form of the strong law.*

A. Foundations and permitted imports

Import PS-01(i): summable event probabilities imply only finitely many occurrences almost surely. Use linearity of expectation, factorization of expectations of independent integrable products, and Markov's inequality for nonnegative Z: $\mathbb{P}(Z > a) \leq \mathbb{E}Z/a$ for $a > 0$, obtained by $Z \geq a1_{\{Z>a\}}$ and integration. Lower moments are finite by Hölder on a probability space. Centering preserves a finite fourth moment via $|x-y|^4 \leq 8(|x|^4 + |y|^4)$.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $\mathbb{E}S_n^4 = nm_4 + 3n(n-1)\sigma^4$.

Receiving activity. N2.1

Compatibility. E1/v1 and any fixed $\varepsilon > 0$. Check: The bound must be summable in n; a variance-based $1/n$ bound is insufficient for this direct argument.

T2-3: N2/E2/v1 to N3. $\sum_n \mathbb{P}(|S_n/n| > \varepsilon) < \infty$.

Receiving activity. N3.1

Compatibility. E2/v1 for each $\varepsilon = 1/r$, $r \in \mathbb{N}$. Check: An uncountable intersection of probability-one events is not justified by countable additivity.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Expand the fourth moment

Object and role. Classify the index multiplicities in the fourth-power expansion and use centering to eliminate singleton factors. **Local obligation:** $\mathbb{E}S_n^4 = nm_4 + 3n(n-1)\sigma^4$.

Inputs. $Y_j = X_j - \mu$, $S_n = \sum_{j=1}^n Y_j$, $\sigma^2 = \mathbb{E}Y_1^2$, and $m_4 = \mathbb{E}Y_1^4$.

C — Contract

Publish E1/v1 only after establishing this obligation: $\mathbb{E}S_n^4 = nm_4 + 3n(n-1)\sigma^4$.

K — Knowledge

$Y_j = X_j - \mu$, $S_n = \sum_{j=1}^n Y_j$, $\sigma^2 = \mathbb{E}Y_1^2$, and $m_4 = \mathbb{E}Y_1^4$. Retain the stated assumptions and import versions.

S — Strategy

Classify the index multiplicities in the fourth-power expansion and use centering to eliminate singleton factors.

V — Validation

Independence and zero means eliminate terms with any index appearing exactly once; pair terms have combinatorial coefficient six.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Expand $(\sum_j Y_j)^4$. If an index occurs once in a term, independence factors out its zero mean, so that term has expectation zero. The surviving terms are the n fourth powers Y_j^4 and terms with two distinct indices each repeated twice.

N1.2 For each unordered pair $i < j$, there are $4!/(2!2!) = 6$ placements, and the expectation of $Y_i^2 Y_j^2$ is σ^4. Thus the total is $nm_4 + 6\binom{n}{2}\sigma^4 = nm_4 + 3n(n-1)\sigma^4$. All terms are integrable by the declared moment and independence.

Export E1/v1. $\mathbb{E}S_n^4 = nm_4 + 3n(n-1)\sigma^4$.

Acceptance gate. Independence and zero means eliminate terms with any index appearing exactly once; pair terms have combinatorial coefficient six.

Targeted repair. If all cross terms are discarded, restore the paired-index terms; centering kills singletons, not products of squares. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Obtain summable deviation probabilities

Object and role. Apply Markov to the fourth power of the centered sample mean. **Local obligation:** $\sum_n \mathbb{P}(|S_n/n| > \varepsilon) < \infty$.

Inputs. E1/v1 and any fixed $\varepsilon > 0$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $\sum_n \mathbb{P}(|S_n/n| > \varepsilon) < \infty$.

K — Knowledge

E1/v1 and any fixed $\varepsilon > 0$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Apply Markov to the fourth power of the centered sample mean.

V — Validation

The bound must be summable in n; a variance-based $1/n$ bound is insufficient for this direct argument.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Markov's inequality and E1 give $\mathbb{P}(|S_n/n| > \varepsilon) \le [nm_4 + 3n(n-1)\sigma^4]/(\varepsilon^4 n^4) \le (m_4 + 3\sigma^4)/(\varepsilon^4 n^2)$.

N2.2 The series $\sum_n n^{-2}$ converges, so these probabilities are summable. PS-01(i) now says that, for this fixed tolerance, $|S_n/n| > \varepsilon$ occurs only finitely often outside a null set.

Export E2/v1. $\sum_n \mathbb{P}(|S_n/n| > \varepsilon) < \infty$.

Acceptance gate. The bound must be summable in n; a variance-based $1/n$ bound is insufficient for this direct argument.

Targeted repair. If the proof applies Borel–Cantelli to a nonsummable upper bound, use the fourth-moment estimate instead of merely a variance estimate. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Synchronize all tolerances

Object and role. Intersect countably many full-probability events so that every required accuracy holds along the same sample path. **Local obligation:** $S_n/n \to 0$ almost surely, equivalently the sample mean converges to μ.

Inputs. E2/v1 for each $\varepsilon = 1/r$, $r \in \mathbb{N}$. Incoming proof exports: E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: $S_n/n \to 0$ almost surely, equivalently the sample mean converges to μ.

K — Knowledge

E2/v1 for each $\varepsilon = 1/r$, $r \in \mathbb{N}$. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Intersect countably many full-probability events so that every required accuracy holds along the same sample path.

V — Validation

An uncountable intersection of probability-one events is not justified by countable additivity.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 For each integer $r \geq 1$, let E_r be the probability-one event on which eventually $|S_n/n| \leq 1/r$. The event $E = \bigcap_{r \geq 1} E_r$ has probability one because its complement is a countable union of null sets.

N3.2 Fix a sample point in E and any $\delta > 0$. Choose r with $1/r < \delta$. The eventual bound on that path yields $|S_n/n| < \delta$ for all sufficiently large n. Thus $S_n/n \to 0$ on E, and adding μ proves the theorem.

Export E3/v1. $S_n/n \to 0$ almost surely, equivalently the sample mean converges to μ.

Acceptance gate. An uncountable intersection of probability-one events is not justified by countable additivity.

Targeted repair. If different null sets are silently ignored for every real tolerance, replace the uncountable family by the countable tolerances $1/r$. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Expand the fourth moment

1. **N1.1.** Expand $(\sum_j Y_j)^4$. If an index occurs once in a term, independence factors out its zero mean, so that term has expectation zero. The surviving terms are the n fourth powers Y_j^4 and terms with two distinct indices each repeated twice.

2. **N1.2.** For each unordered pair $i < j$, there are $4!/(2!2!) = 6$ placements, and the expectation of $Y_i^2 Y_j^2$ is σ^4. Thus the total is $nm_4 + 6\binom{n}{2}\sigma^4 = nm_4 + 3n(n-1)\sigma^4$. All terms are integrable by the declared moment and independence.

B. N2 — Obtain summable deviation probabilities

1. **N2.1.** Markov's inequality and E1 give $\mathbb{P}(|S_n/n| > \varepsilon) \le [nm_4 + 3n(n-1)\sigma^4]/(\varepsilon^4 n^4) \le (m_4 + 3\sigma^4)/(\varepsilon^4 n^2)$.

2. **N2.2.** The series $\sum_n n^{-2}$ converges, so these probabilities are summable. PS-01(i) now says that, for this fixed tolerance, $|S_n/n| > \varepsilon$ occurs only finitely often outside a null set.

C. N3 — Synchronize all tolerances

1. **N3.1.** For each integer $r \ge 1$, let E_r be the probability-one event on which eventually $|S_n/n| \le 1/r$. The event $E = \bigcap_{r \ge 1} E_r$ has probability one because its complement is a countable union of null sets.

2. **N3.2.** Fix a sample point in E and any $\delta > 0$. Choose r with $1/r < \delta$. The eventual bound on that path yields $|S_n/n| < \delta$ for all sufficiently large n. Thus $S_n/n \to 0$ on E, and adding μ proves the theorem.

□

V. WORKED ORCHESTRATION

For IID Bernoulli(1/2) variables, $\mu = 1/2$, $\sigma^2 = 1/4$, and $m_4 = 1/16$. Thus $\mathbb{E}S_n^4 = (3n^2 - 2n)/16$, and $\mathbb{P}(|n^{-1}\sum X_j - 1/2| > \varepsilon) \le 3/(16\varepsilon^4 n^2)$. Summability proves that the empirical proportion converges almost surely to 1/2.

A. Boundary and hypothesis audit

If all variables are the same nonconstant Bernoulli variable Z, they are identically distributed with every moment finite, but their sample mean is always Z, not its mean 1/2.

Independence is essential to this theorem. Conversely failure of a fourth moment does not itself refute a strong law; it only falls outside this proved version.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 shortcut: Chebyshev gives C/n, so Borel–Cantelli proves almost sure convergence. The series of those upper bounds diverges, leaving its hypothesis unproved. **Repair:** under the stated fourth-moment assumption, use E1 to obtain the summable C_ε/n^2 bound. This repairs the proof without strengthening the stated theorem. Recheck the countable tolerance intersection afterward.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Prove the optimal IID strong law under a finite first moment using truncation and a different summability argument; do not label this restricted proof as that stronger theorem.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Amir Dembo, Probability Theory: STAT310/MATH230 notes for measure-theoretic probability and limit arguments; Stanford STAT310C syllabus and lecture materials for martingale inequalities and stopping theorems. The versions and proofs used in this collection are stated explicitly.